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KCMamba: Kalman-Guided Selective State-Space Model for Robust Time Series Forecasting

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A szakdolgozat címe: Kalman-alapú állapotter-modelle fejlesztése idősoros becsléshez

A szakdolgozat témája:

(A témavezetővel konzultálva adja meg 1/2 - 1 oldal terjedelemben szakdolgozat témájának leírását)

Motiváció

Számos alkalmazási terület – járműkövetés, robot szenzorfüzió, szenzorhálózatok és energia/pénzügyi előrejelzés – zajos, nagyméretű idősorok valós idejű feldolgozását igényli. A gyakorlatban alkalmazott módszerek gyakran számítási szempontból drágák, érzékenyek a mérési anomáliákra, és nehezen skálázhatók hosszú sorozatokra és magas állapotdimenziókra. A szakdolgozat célja egy Kalman-szűrőn alapuló állapotter-modelle megtervezése és kiértékelése.

Célkitűzések

1. Kalman-alapú modelle megtervezése SSM vagy Mamba modellek alapján.
2. Működés kiértékelése ilyen feltételek mellett.
3. Az új megközelítés empirikus összehasonlítása a standard alapvonallakkal több, eltérő adathalmazon.

Kutatási kérdések

- A bizonytalanságbecslés közvetlenül iterálható-e egy strukturált állapotter-modellebe, és milyen mértékben előnyös ez?
- Hogyan viselkedik a modelle különböző zajszintek, részleges szenzorhibák és kiugró értékek esetén?
- Mennyire általánosítható a megközelítés különböző területeken (követés, szenzorfüzió, pénzügyi/energia idősorok)?

Módszertani megközelítés

A szakdolgozat egy Kalman-családba tartozó, állapotter-becslőt épít.

A fejlesztés a Kalman-szűrőben található bizonytalanságbecslés és predikció-frissítés hurkok integrálására összpontosít a Strukturált Állapotter-modellek központi blokkjába.

Értékelési terv és metrikák

A modelle több adathalmazon tipikus alapértékekkel hasonlítjuk össze. A főbb metrikák közé tartozik az RMSE/MAE (becslési hiba), a negatív log-valószínűség vagy a kalibrációs metrikák (bizonytalanságkezelés). A dedikált ablációs vizsgálatok számszerűsítik, hogy az egyes komponensek hogyan járulnak hozzá a sebesség- és stabilitásnövekedéshez.

Budapest, 2025. 10. 15.

Abstract

This thesis presents **KCMamba** (Kalman Core-Attention Mamba), a hybrid neural architecture that integrates a differentiable Kalman filtering mechanism into the Mamba selective state-space model (SSM) for noise-robust time series forecasting. The proposed architecture employs learned Q_{net} , R_{net} , and K_{net} subnetworks that adaptively regulate the ratio of process and measurement noise, enabling the model to behave as a passive predictor on clean data and as an active filter under noisy conditions.

We conduct empirical evaluation on three public LTSF benchmarks (Exchange-Rate, ETTh1, ETTh2) under synthetically injected noise ($\sigma \in \{0, 1, 3, 5\}$) and low data-availability regimes (stride $\in \{1, 16\}$). Our results show that KCMamba achieves an average MSE degradation of 1528% as noise increases from $\sigma = 0$ to $\sigma = 5$, compared to 28234% for Fair Mamba, a $18.5\times$ robustness advantage.

The thesis also describes the complete production software system: an asynchronous, event-driven trading engine built in a hexagonal architecture, a web-based dashboard, a backtesting framework, and a live Binance trading adapter.

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Chapter 1

Introduction

Time series forecasting is one of the most studied areas of applied machine learning, partly because real-world data is almost always noisy. Sensor dropouts, aggregation artefacts, and market microstructure noise all degrade the signal-to-noise ratio, and models frequently struggle with this. In the long-term forecasting (LTSF) setting the problem is harder still, since a lookback window of length L must support a prediction of the next H steps, where H can exceed L [1].

1.1 Motivation

Transformer-based models [1, 2] and the Mamba SSM [3, 4] address data-quality issues through implicit generalisation, relying on the model to learn around the noise. This works on clean data, but there is no guarantee that the robustness transfers to noisy regimes.

The Kalman filter [5] takes a different route: it builds an *explicit* model of process noise (Q) and measurement noise (R), and derives a gain K from their ratio to decide how much to trust the current measurement versus the prior state. The **KCMamba** architecture presented in this thesis is built on this principle. That inductive bias, if embedded in a neural network, may help in noisy conditions.

1.2 Research Questions

Three questions are investigated:

1. Is KCMamba competitive on clean data against standard LTSF models?

2. How much MSE degradation do models suffer under synthetically injected noise ($\sigma \in \{0, 1, 3, 5\}$)?
3. Which architecture generalises better under scarce training data (stride $\in \{1, 16\}$)?

1.3 Contributions

- A description of the **KCMamba** architecture: learned Q_{net} , R_{net} , K_{net} sub-networks inside a differentiable Kalman filter, with a parallel prefix-scan GPU implementation.
- Measurements showing that the learned K adapts to noise level: $K \approx 0.71$ at $\sigma = 0$ (passive pass-through) and $K \approx 0.07$ at $\sigma = 5$ (strong filtering).
- Benchmark results on three datasets (Exchange-Rate, ETTh1, ETTh2): Fair Mamba degrades by 28 234% under noise, whereas KCMamba degrades by only 1 528% ($18.5\times$ smaller degradation).
- Long-sequence analysis (ETTm1, Exchange, $T \in \{96, 192, 336, 512\}$): KCMamba achieves **31.7%** lower MSE than LSTM at $T=336$, $\sigma=1$ (0.502 vs 0.735). At $T=512$, $\sigma=3$, KCMamba scores 0.773 while Fair Mamba scores 2.121 (63.6% better). On ETTm1 at $\sigma=3$, KCMamba is the only model whose MSE monotonically decreases with sequence length ($0.673 \rightarrow 0.466$ as $T: 96 \rightarrow 512$, -31%).
- Stride $\times$ sequence-length sweep (Exchange, $s \in \{1, 16\}$, $T \in \{192, 336, 512\}$): at $s=1$, $\sigma=3$, KCMamba is **+35%** better than LSTM at $T=512$ (0.836 vs 1.292), and the advantage grows monotonically with sequence length. At $s=16$, $\sigma \geq 3$, Fair Mamba’s MSE rises to 8.1–8.5 while KCMamba stays at 1.1–1.3.
- An open-source Python/PyTorch/CUDA implementation and an asynchronous trading system for deploying the model in live markets.

1.4 Thesis Structure

Chapter 2 reviews the Kalman filter, SSMs, and the LTSF literature. Chapter 3 describes the KCMamba architecture and its mathematical underpin-

nings. Chapter 4 presents the experimental setup and results. Chapter 5 covers the implemented software system. Chapter 7 draws conclusions and discusses future work.

Chapter 2

Related Work

2.1 The Kalman Filter and Its Extensions

The Kalman filter [5] is the minimum-variance, unbiased, linear estimator for systems whose state $x_t \in \mathbb{R}^n$ evolves according to a linear dynamical model perturbed by Gaussian noise:

$$x_t = Fx_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q), \quad (2.1)$$

$$y_t = Hx_t + v_t, \quad v_t \sim \mathcal{N}(0, R), \quad (2.2)$$

where F is the state-transition matrix, H the observation matrix, $Q \in \mathbb{R}^{n \times n}$ the process-noise covariance, and $R \in \mathbb{R}^{m \times m}$ the measurement-noise covariance. The filter operates in two alternating half-steps. In the prediction step (time update), before the measurement y_t arrives, the filter propagates the state estimate and the error covariance forward in time:

$$\hat{x}_{t|t-1} = F\hat{x}_{t-1|t-1}, \quad (2.3)$$

$$P_{t|t-1} = FP_{t-1|t-1}F^\top + Q, \quad (2.4)$$

where $P_{t|t-1} \in \mathbb{R}^{n \times n}$ is the predicted error covariance. A large $P_{t|t-1}$ indicates high prediction uncertainty; a large Q means the system state changes rapidly and unpredictably.

In the correction step (measurement update), once the observation y_t arrives, the filter corrects its estimate by the *innovation* $e_t = y_t - H\hat{x}_{t|t-1}$, weighted by the

Kalman gain K_t :

$$K_t = P_{t|t-1} H^\top (H P_{t|t-1} H^\top + R)^{-1}, \quad (2.5)$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t e_t, \quad (2.6)$$

$$P_{t|t} = (I - K_t H) P_{t|t-1}. \quad (2.7)$$

Equation (2.5) encodes the fundamental noise-balance intuition. When the measurement noise R is large relative to the predicted covariance $H P_{t|t-1} H^\top$, the denominator dominates and $K_t \rightarrow 0$: the filter distrusts the observation and relies on its prior state. When $R \rightarrow 0$, $K_t \rightarrow H^{-1}$ and the observation nearly overwrites the state entirely.

For the scalar case ($F = H = 1$), the gain simplifies to

$$K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R}. \quad (2.8)$$

Under steady-state conditions the error covariance $P_{t|t-1}$ converges to a fixed value $\bar{P}$, and the steady-state gain depends only on the ratio Q/R : a high process noise relative to measurement noise drives K towards 1 (trust the measurement), while a low Q/R ratio drives K towards 0 (trust the model). This $Q/(Q + R)$ ratio is the structural basis for the KCMamba gain computation (Section 3.2.3).

2.1.1 Learned Kalman Filters

In the neural Kalman literature, the matrices F, H, Q, R are parameterised by neural networks [6, 7]. The *Deep Kalman Filter* [6] allows non-linear transitions, and the *Kalman Variational Autoencoder* [7] learns disentangled representations of latent states. KCMamba builds on these ideas but operates as a sequence forecaster rather than as an autoencoder.

2.2 State Space Models

2.2.1 S4

S4 [3] reformulates a sequence model as a parameterised ordinary differential equation (ODE) over a latent state $h(t) \in \mathbb{R}^N$:

$$h'(t) = Ah(t) + Bu(t), \quad (2.9)$$

$$y(t) = Ch(t). \quad (2.10)$$

Here $u(t)$ is the scalar input, $A \in \mathbb{R}^{N \times N}$ is the state matrix, and $B, C \in \mathbb{R}^N$ are the input and output projections. The critical insight of Gu et al. is that A can be initialised as the *HiPPO* (High-order Polynomial Projection Operator) matrix, which is designed to compress the history of u into the current state $h(t)$ with optimal polynomial approximation. This initialisation avoids the vanishing-gradient problem that plagues vanilla RNNs over long sequences, since the HiPPO structure preserves information from the distant past without exponential decay.

To apply the model to discrete sequences the continuous system is discretised with step size Δ using the zero-order hold (ZOH) rule:

$$h_t = \bar{A}h_{t-1} + \bar{B}u_t, \quad (2.11)$$

$$y_t = Ch_t, \quad (2.12)$$

where $\bar{A} = e^{\Delta A}$ and $\bar{B} = (e^{\Delta A} - I)A^{-1}B$. Because $\bar{A}$ is diagonal in the frequency domain the entire convolution $y = \bar{C} * u$ can be computed as a single element-wise product after a Fast Fourier Transform (FFT), reducing sequence-level complexity to $O(T \log T)$.

A fundamental limitation of S4 is that A , B , C , and Δ are *input-independent*: the same learned parameters are applied at every time step regardless of the input content. The model therefore cannot adapt its memory or forgetting behaviour in response to what it actually observes in the current sequence.

2.2.2 Mamba

Mamba [4] addresses this limitation directly by making the state-space parameters *input-dependent*. The matrices B , C , and the time step Δ are no longer fixed

scalars but functions of the current input u_t :

$$B_t = \text{Linear}(u_t), \quad (2.13)$$

$$C_t = \text{Linear}(u_t), \quad (2.14)$$

$$\Delta_t = \text{softplus}(\text{Linear}(u_t) + \delta_{\text{bias}}). \quad (2.15)$$

The time step Δ_t determines the discrete transition matrices $\bar{A}_t = e^{\Delta_t A}$, $\bar{B}_t = (\bar{A}_t - I)A^{-1}B_t$. This mechanism, called the *Selective State Space* (S6) layer, lets the model selectively remember or forget content at each step. When $\Delta_t \rightarrow 0$ the state barely changes ($\bar{A}_t \approx I$), preserving memory. When Δ_t is large the state is updated strongly, discarding older information.

The input-dependent parameters break the global convolution trick that made S4 efficient, because there is no longer a single kernel to transform via FFT. Gu and Dao solve this with a *hardware-efficient parallel scan* (the Blelloch prefix-scan algorithm), applied on the GPU’s SRAM to avoid expensive high-bandwidth memory round-trips. The scan runs in $O(\log T)$ synchronisation steps on T parallel threads, achieving $O(T)$ total work and $O(T)$ memory while avoiding the sequential bottleneck of recurrent models.

2.2.3 Fair Mamba

The comparison baseline, referred to as “Fair Mamba”, is a channel-wise Mamba with identical hyperparameters and training protocol as KCMamba. The “fair” label means the parameter budget and data are identical, so any difference in results comes solely from the architecture.

2.3 Long-Term Time Series Forecasting

2.3.1 Problem Formulation

Long-term time series forecasting (LTSF) considers an input window $\mathbf{X} = (x_1, \dots, x_L) \in \mathbb{R}^{L \times d}$ of L historical steps across d channels, and requires producing a prediction $\hat{\mathbf{X}} = (\hat{x}_{L+1}, \dots, \hat{x}_{L+H}) \in \mathbb{R}^{H \times d}$ for the next H steps. The objective

is to minimise the normalised mean squared error

$$\text{MSE} = \frac{1}{Hd} \sum_{h=1}^H \sum_{j=1}^d \frac{(x_{L+h,j} - \hat{x}_{L+h,j})^2}{\text{Var}(x_{:,j})}, \quad (2.16)$$

where normalisation by the per-channel variance makes results comparable across datasets with different scales. The *long-term* qualifier refers to the common regime $H \geq L$ or H comparable to L : the model must extrapolate well beyond the lookback window, which is substantially harder than short-term forecasting where $H \ll L$.

2.3.2 Reference Models

Informer [1] was the first Transformer variant specifically designed for long-term forecasting. Standard self-attention has $O(L^2)$ time and memory complexity, which becomes prohibitive for long input windows. Informer addresses this with *ProbSparse attention*: it identifies the $O(\log L)$ most informative query positions via a KL-divergence estimate, reducing complexity to $O(L \log L)$. A distilling operation halves the sequence length between encoder layers so the model can handle lookback windows of several thousand steps.

Autoformer [2] decomposes each series into a trend component (estimated with a moving-average kernel) and a seasonal residual, applying the Transformer only to the residuals. Rather than dot-product attention it uses an *Auto-Correlation* mechanism: it identifies the lag τ maximising the autocorrelation of the query signal and aggregates values at those lag offsets. The resulting mechanism is $O(L \log L)$ via FFT and provides an inductive bias well suited to periodic data.

DLinear [8] discards all of this complexity and applies a single linear layer to each lookback window to produce the forecast, optionally after trend-seasonal decomposition. Despite its simplicity, DLinear matched or outperformed many Transformer variants on standard benchmarks, demonstrating that architectural complexity is not always necessary for good forecasting performance. It remains a relevant baseline because any new model must justify its additional overhead relative to this linear reference.

2.3.3 Datasets

The Exchange-Rate dataset [9] contains eight foreign exchange rates sampled daily from 1990 to 2016, giving 7 588 time steps per channel. Exchange-rate data has near-unit-root dynamics: each daily increment is close to a random walk with very little mean reversion. This makes it a challenging benchmark for autoregressive models, and a natural setting for evaluating the Kalman structure where the signal-to-noise ratio is intrinsically low.

ETTh1 and ETTh2 [1] are both collected from Chinese transformer stations, recording hourly electricity load and oil temperature for two years (17 420 steps, $d = 7$ channels). ETTh1 has a clear daily periodicity, while ETTh2 is noisier and more irregular. The two datasets provide contrasting difficulty levels for noise-robustness evaluation.

2.4 Noise Robustness

The noise sensitivity of time series models is rarely studied in a systematic way. PatchTST [10] indirectly improves noise tolerance through patch tokenisation, but robustness is not its primary goal.

In this work the evaluation protocol is: $\tilde{x}_t = x_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma \in \{0, 1, 3, 5\}$. Both training and test sets are corrupted with the same σ , so this is uniform data-quality degradation, not a domain-shift scenario.

Chapter 3

The KCMamba Architecture

3.1 Overview

KCMamba (Kalman Core-Attention Mamba) is a hybrid neural time-series model that embeds a differentiable, content-dependent Kalman filter layer into the Mamba SSM [4]. Instead of using a fixed forgetting factor A , the model learns a gain K from the current input and uses it to decide how much to trust the new measurement versus the prior state.

The model has two levels: the base processing unit is a `KCMambaBlock` (content-dependent Kalman filtering and state update), stacked into a `KCMambaStack` of N sequential blocks with a cross-attention layer at the top.

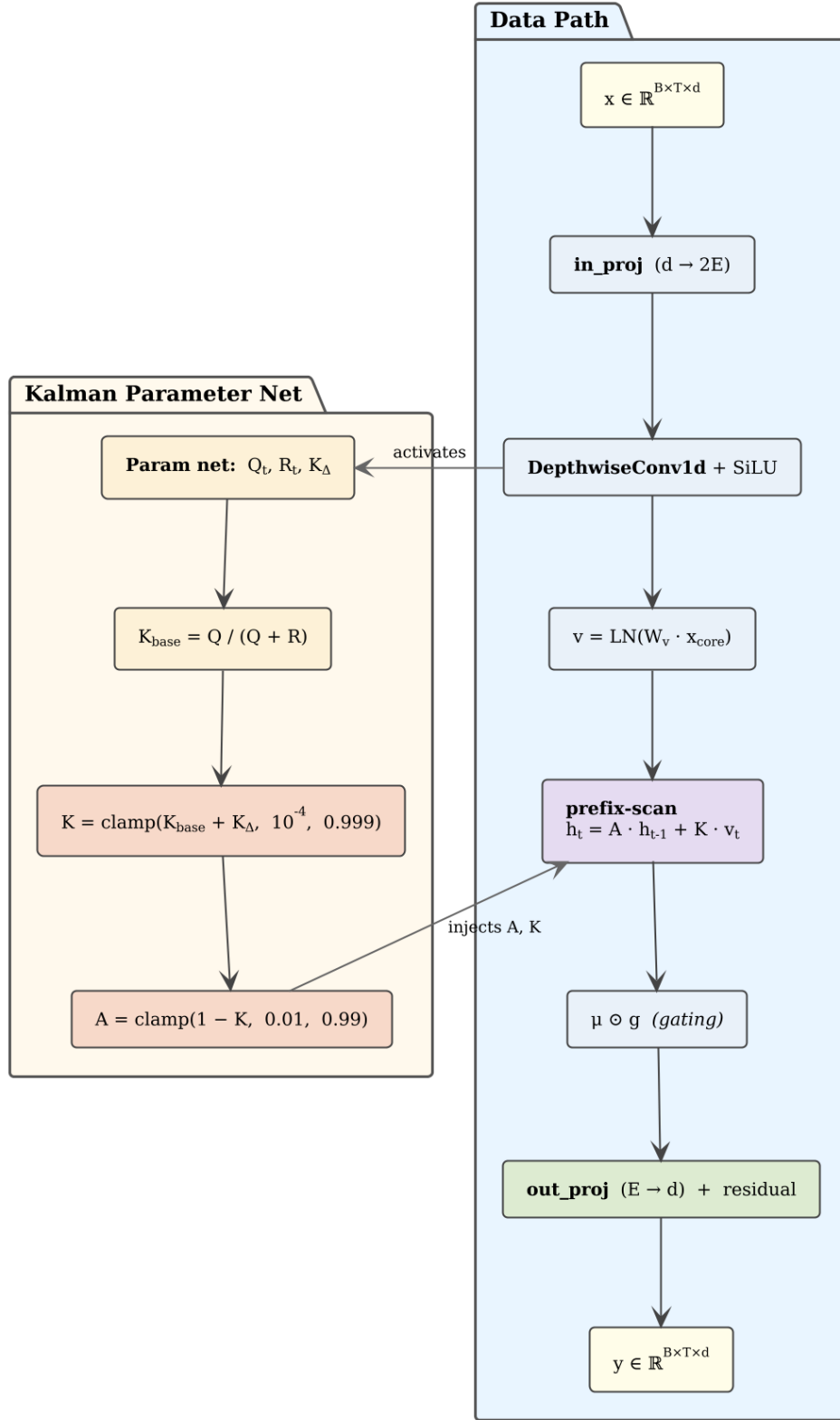


Figure 3.1: Full data-flow diagram of the KCMamba block. Yellow blocks (Q_{net} , R_{net} , K_{net}) learn the Kalman statistics. The purple block performs the parallel prefix-scan on GPU in $O(\log T)$ synchronisation steps, and green blocks denote input/output projections.

3.2 KCMamba Block

Let the input sequence be $x \in \mathbb{R}^{B \times T \times d}$, where B is the batch size, T is the sequence length, and d is the feature dimension.

3.2.1 Input Projection and 1D Convolution

The input is projected into two streams $(x_{\text{cw}}, x_{\text{gate}})$ by a linear layer $W_{\text{in}} \in \mathbb{R}^{2d_{\text{inner}} \times d}$, followed by SiLU gating:

$$[x_{\text{cw}}, x_{\text{gate}}] = W_{\text{in}}x, \quad (3.1)$$

$$\text{gate} = \text{SiLU}(x_{\text{gate}}). \quad (3.2)$$

The dual-stream design separates two responsibilities. x_{cw} carries the signal to be filtered by the Kalman layer, while x_{gate} learns an independent amplitude modulation applied at the output stage. This prevents the gating signal from interfering with the Kalman state dynamics. $\text{SiLU}(z) = z \cdot \sigma(z)$ is preferred over ReLU because it is smooth everywhere and allows small negative outputs, which benefits gradient flow during early training.

Before the global sequential scan, a depthwise 1D convolution ($k = 4$ kernel) on x_{cw} integrates short-range temporal context:

$$x_{\text{core}} = \text{SiLU}(\text{Conv1D}(x_{\text{cw}})). \quad (3.3)$$

Operating channel-by-channel (depthwise), the convolution adds only $k \cdot d_{\text{inner}}$ extra parameters while capturing local temporal patterns such as short-term momentum and candlestick-level trend direction, before the Kalman filter processes the full sequence.

3.2.2 Kalman Parameter Networks

Three small two-layer fully connected MLP sub-networks learn the Kalman filter statistics. Q_{net} and R_{net} produce the per-step noise variance estimates:

$$Q_t = \text{softplus}(Q_{\text{net}}(x_{\text{core}})) \cdot \text{softplus}(q_{\text{scale}}), \quad (3.4)$$

$$R_t = \text{softplus}(R_{\text{net}}(x_{\text{core}})) \cdot \text{softplus}(r_{\text{scale}}). \quad (3.5)$$

$Q_t > 0$ denotes the *process-noise* estimate. When Q_t is large, the hidden state is expected to change rapidly, so the current measurement receives more weight. $R_t > 0$ denotes the *measurement-noise* estimate. A large R_t indicates a noisy observation, causing the model to rely more on its prior state. The softplus activation, defined as $\text{softplus}(z) = \ln(1 + e^z) > 0$, guarantees strict positivity of both quantities, as required by the Kalman derivation. The learnable scalars q_{scale} and r_{scale} set a global regime bias on top of the per-step outputs. Initialising $q_{\text{scale}} = -2.0$ and $r_{\text{scale}} = 1.5$ gives $R \gg Q$ at the start, corresponding to $K_{\text{base}} \approx 0.07$, meaning strong filtering by default.

A third sub-network, K_{net} , is also a two-layer fully connected Multi-Layer Perceptron that provides a direct residual correction to the final gain, described in Section 3.2.3. The $Q_{\text{net}}/R_{\text{net}}$ pair models the slowly evolving noise regime of the time series, whereas K_{net} can react immediately to fast, content-specific events such as sudden volatility spikes or trend reversals that the noise-ratio estimate has not yet adapted to.

3.2.3 Kalman Gain Computation

The base Kalman gain is derived from the $Q/(Q + R)$ noise ratio, which is the classical steady-state formula from estimation theory:

$$K_{\text{base},t} = \frac{Q_t}{Q_t + R_t + \varepsilon}, \quad (3.6)$$

where $\varepsilon = 10^{-6}$ prevents division by zero. When $Q \gg R$, meaning the state changes rapidly, $K \rightarrow 1$ and the model follows the current observation closely; when $R \gg Q$, meaning the observations are noisy, $K \rightarrow 0$ and the prior state dominates. This provides an interpretable, physics-grounded prior that aligns with classical control theory.

A content-dependent correction from K_{net} is added to handle rapid regime changes that the slowly-adapting Q/R ratio may not immediately capture:

$$K_t = \text{clamp}(K_{\text{base},t} + 0.3(\sigma(K_{\text{net}}(x_{\text{core}})) - 0.5), [10^{-4}, 0.999]). \quad (3.7)$$

The -0.5 shift centres the correction at zero at initialisation, so early training starts from the Q/R prior rather than a random offset. The clamp to $[10^{-4}, 0.999]$ guards

against degenerate extremes. $K_t \approx 0$ would freeze state updates with no new information integrated, while $K_t \approx 1$ would discard all memory from the previous step.

3.2.4 State Update with Parallel Scan

With forgetting factor $A_t = \text{clamp}(1 - K_t, 0.01, 0.99)$, the discrete Kalman recurrence is:

$$\begin{aligned} h_t &= A_t \odot h_{t-1} + K_t \odot v_t, \\ v_t &= \text{LayerNorm}(W_v x_{\text{core},t}). \end{aligned} \tag{3.8}$$

where $A_t \in (0, 1)^{d_{\text{inner}}}$ is the forgetting factor vector, with $A_t \rightarrow 1$ indicating strong memory and $A_t \rightarrow 0$ indicating a strong update, $\odot$ is the element-wise Hadamard product, $h_{t-1} \in \mathbb{R}^{d_{\text{inner}}}$ is the previous step's latent state, v_t is the current value vector, and $W_v \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{inner}}}$ is a learnable projection.

Efficient evaluation uses the Blelloch parallel prefix-scan [11]:

$$A_{\text{new}}[t] \leftarrow A[t] \cdot A[t - \text{step}], \tag{3.9}$$

$$B_{\text{new}}[t] \leftarrow B[t] + A_{\text{orig}}[t] \cdot B[t - \text{step}], \tag{3.10}$$

where $A[t]$ and $B[t]$ are the associative prefix-pair (a_t, b_t) : a_t is the accumulated forgetting factor, b_t is the accumulated input sum. These are *not* the same as the SSM matrices A , B . The stride $\text{step} \in \{1, 2, 4, \dots, T/2\}$ doubles each pass. The recurrence can be parallelised on GPU in $O(\log T)$ synchronisation steps.

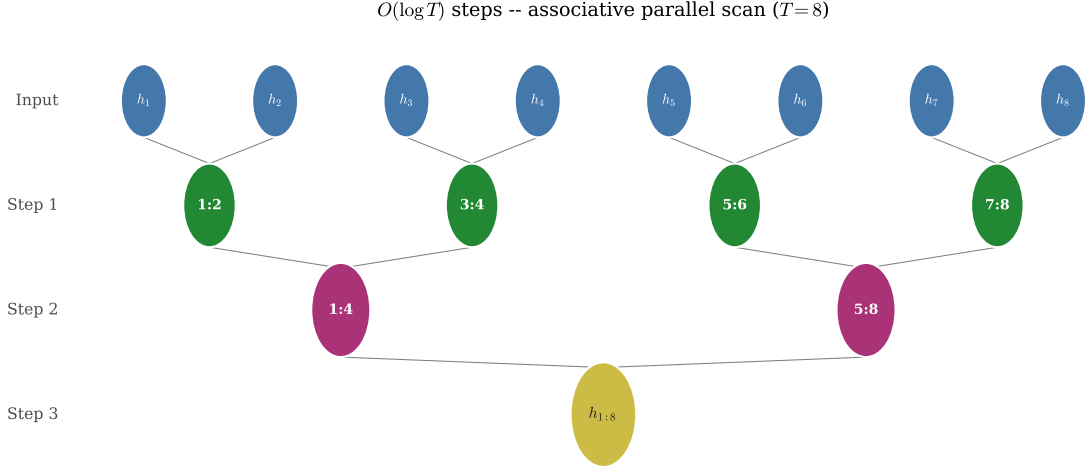


Figure 3.2: Blleloch parallel prefix-scan on a length-8 sequence. Each pass combines two elements separated by the current step size, and after $\log_2 T = 3$ passes every h_t contains the full context. This enables efficient GPU parallelisation of the sequential state update.

3.2.5 Output Projection and Residual Connection

The Kalman state sequence (μ) is combined with the gate:

$$y_{\text{KC}} = W_{\text{out}}(\mu \odot \text{gate}). \quad (3.11)$$

A scalar-gated residual connection preserves the gradient path:

$$\text{out} = y_{\text{KC}} + \sigma(b_{\text{res}}) \cdot (W_{\text{res}} x - y_{\text{KC}}), \quad (3.12)$$

where b_{res} is a learnable scalar initialised at -1.0 , giving $\sigma(b_{\text{res}}) \approx 0.27$, so the SSM path dominates by default.

3.3 KCMamba Stack

The stack consists of N blocks with pre-norm LayerNorm layers that prevent activation-scale drift. A *multi-scale cross-attention* layer is appended after the final block. The reason is that Mamba’s recurrent dynamics encode recent time steps more strongly than distant ones in the hidden state. For long-horizon forecasting, however, slow trends must also be accessible. The cross-attention queries the current state against the subsampled key/value set, so both short and long time scales are

taken into account:

$$\text{out}[T] = h[T] + \text{MHA}(h[T], h[:, s], h[:, s]), \quad (3.13)$$

where $h[T] \in \mathbb{R}^{1 \times d}$ is the last time step's representation, serving as the query; $h[:, s] \in \mathbb{R}^{[T/s] \times d}$ is the every- s -th subsampled sequence, representing the slower, coarser time scale as keys and values; and $\text{MHA}(\text{query}, \text{key}, \text{value})$ denotes multi-head cross-attention in the standard sense [12].

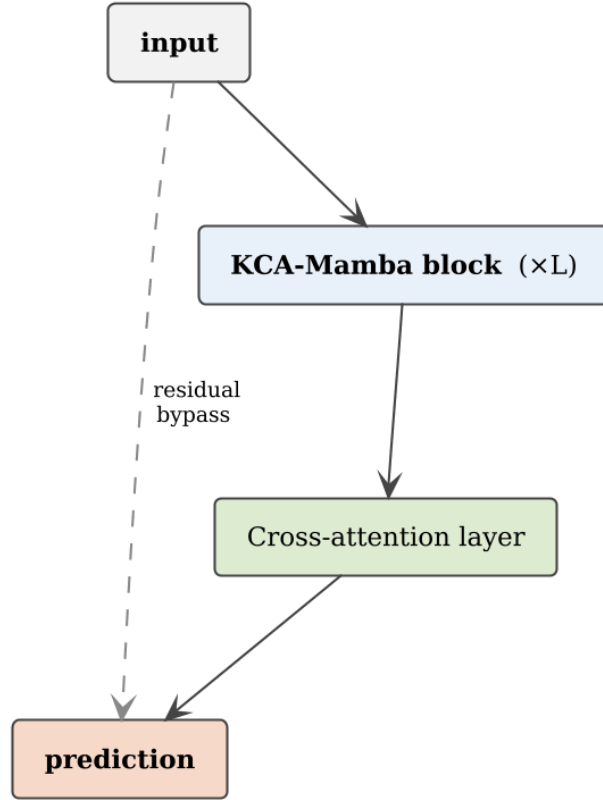


Figure 3.3: KCMamba stack structure. From input to output: input projection, 2 KC-blocks with LayerNorm, output normalisation, and cross-attention for the slow ($s = 4$) time scale.

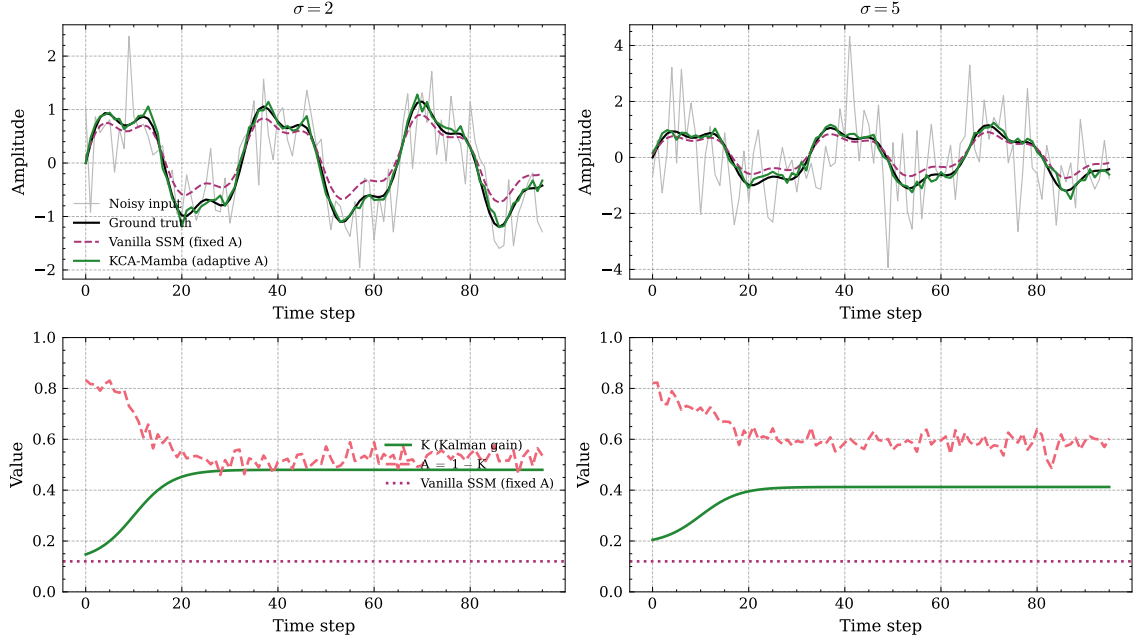


Figure 3.4: Synthetic signal filtering: **vanilla SSM** (standard Mamba, fixed A) vs. **KCMamba** (adaptive K -gain) at moderate ($\sigma = 2$) and strong ($\sigma = 5$) noise. *Top rows*: predicted vs. true signal, where KCMamba tracks more tightly even under noise. *Bottom rows*: learned K values over time. Vanilla SSM has a fixed A , whereas KCMamba’s A adapts with noise level.

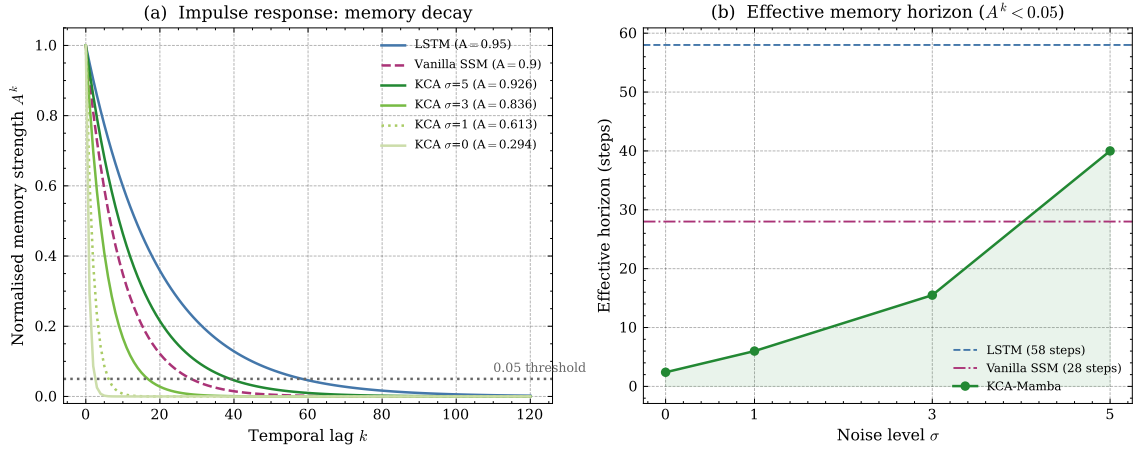


Figure 3.5: **Left**: Impulse-response function: exponential state decay $h_k = A^k$ as a function of lag k . KCMamba’s A grows adaptively with noise ($0.294 \rightarrow 0.926$), extending the memory horizon. **Right**: Effective memory horizon (number of steps until $A^k > 5\%$) vs. noise level. LSTM and vanilla SSM have fixed horizons, whereas KCMamba automatically extends memory under noise.

3.4 Parameter Complexity

Model	Parameters (k)	Layers	State size
LSTM	71.7	1	128
Fair Mamba	53.2	2	H=4, D=32
KCMamba	46.1	2	H=4, D=32

Table 3.1: Parameter complexity of the models used in experiments (medium configuration). KCMamba has fewer parameters than Fair Mamba because the $Q_{\text{net}}/R_{\text{net}}/K_{\text{net}}$ sub-networks redirect resources from the SSM blocks.

3.5 Comparison with Standard Mamba

The key architectural difference between **standard (vanilla) SSM** and KCMamba lies in how the forgetting factor A is handled. Standard Mamba [4] learns A via the time-step Δ , but this process does not separate signal from noise: Δ must model both data structure *and* noise simultaneously.

KCMamba replaces this with an *explicit* Q/R noise domain model: the $K = Q/(Q + R)$ ratio determines whether to trust the current measurement or the prior state. At high noise ($\sigma = 5$), $K \rightarrow 0.073$ and $A = 0.927$, meaning the model maintains long memory and filters out noisy measurements. At low noise ($\sigma = 0$), $K = 0.706$, so the model follows the current data.

This is demonstrated on the synthetic test in Fig. 3.4: standard SSM uses fixed $A = 0.88$ regardless of noise level, while KCMamba adapts A dynamically (Fig. 3.5).

3.6 System Architecture

KCMamba is embedded in a hexagonally structured trading system. Fig. 3.6 shows the main components:

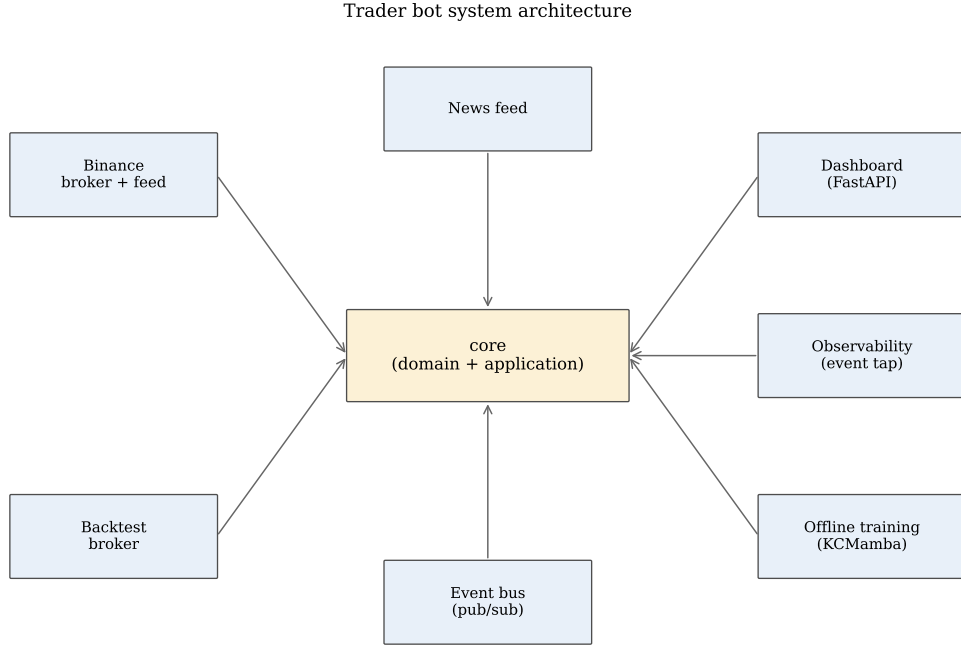


Figure 3.6: Hexagonal architecture of the KCMamba trading system. The **Core** package contains the TradingEngine (pipeline: state estimation $\rightarrow$ prediction $\rightarrow$ signal $\rightarrow$ risk $\rightarrow$ execution), domain models, and application ports. External adapters (Infrastructure, Web, Testing, Offline Training, Observability, Analytics) communicate with the Core exclusively through interfaces (`IEventBusPort`, `IBrokerGatewayPort`, etc.).

3.7 Module and Class Structure

The source code is split into four main packages. `core/` contains all business logic and holds no references to any adapter. Code in `adapters/` accesses the core only through the `ports.py` interfaces.

Package	Module / class	Responsibility
core/domain/	models.py strategy.py risk.py events.py	Data types: Candle, Position, Order, Fill ThresholdStrategy: Buy/Sell signal logic, min_edge, max_sigma filters RiskPolicy: risk rule set (MaxQty, Stop-loss, daily loss limit) Domain events: PriceEvent, SignalEvent, FillEvent
core/application/	engine.py execution.py ports.py stages.py stores.py	TradingEngine: main pipeline orchestrator, internal state machine for the trading cycle Order assembly, sizing, and dispatch via IBrokerGatewayPort Abstract interfaces: IEventBusPort, IBrokerGatewayPort, IMarketFeedPort, ITimeSourcePort Pipeline stages: PredictionStage, SignalStage, RiskStage, ExecutionStage In-memory state: PositionStore, SignalStore, EquityStore
core/ml/	kc_mamba.py mamba_kalman.py services.py feature_engineering.py	KCMambaModel: PyTorch module, forward(x) → forecast KCMambaBlock: Kalman filter layer, parallel prefix-scan MLService: model loading, online update, forecast cache OHLCV → feature vector transform, normalisation, sliding window
adapters/	binance_broker.py binance_feed.py event_bus.py broker.py (testing) dashboard.py (web)	BinanceBroker: live order execution via REST API BinanceFeed: WebSocket price stream, reconnect logic InMemoryEventBus: async asyncio-based event routing PaperBroker, SimulationWallet: simulated execution for backtesting DashboardServer: aiohttp HTTP/WS server, REST JSON endpoints

Table 3.2: Module and class structure of the system. Each class depends only on the data types and interfaces of its own package, while cross-package dependencies are mediated by the `ports.py` abstractions.

Chapter 4

Experiments and Evaluation

4.1 Experimental Setup

4.1.1 Hardware and Software

Training environment: NVIDIA GeForce RTX 3060 (12 GB VRAM), PyTorch 2.x, CUDA 12, float32 precision.

4.1.2 Datasets

Three datasets are used (see also Sec. 2.3.3 in Chapter 2):

- **Exchange-Rate**: 8 currency exchange rates, 1990–2016 (7 588 days, 8 variables) [9].
- **ETTh1**: Chinese transformer-station hourly load and temperature (17 420 steps, 7 variables) [1].
- **ETTh2**: same station, different channel sequence (17 420 steps, 7 variables) [1].

4.1.3 Noise Injection

Noise is added as $\hat{x} = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma \in \{0, 1, 3, 5\}$. Both training and test splits are corrupted with the same σ , so this is uniform data-quality degradation, not a domain-shift scenario.

4.1.4 Model Configuration and Training Hyperparameters

Hyperparameter	LSTM	Fair Mamba	KCMamba
Epochs	5	5	5
Optimiser	Adam	Adam	Adam
Learning rate	3e-3	3e-3	3e-3
Batch size	32	32	32
Sequence length	96	96	96
Forecast horizon	96	96	96
Hidden dim (d)	128	128	128
Heads (H)	n.a.	4	4
State size (D)	n.a.	32	32
Layers	1	2	2
Kernel	n.a.	4	4
Parameters	71.0k	53.2k	46.1k

Table 4.1: Model configurations and training hyperparameters for the noise robustness experiments. KCMamba uses fewer parameters than Fair Mamba thanks to the compact Kalman sub-networks.

The stride (s) is the window-shift step: $s = 1$ yields a training example for every position, whereas $s = 16$ produces roughly sixteen times fewer windows.

4.2 Results

4.2.1 Exchange-Rate Dataset

Stride	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$
$s = 1$	Fair	0.02	0.38	0.98	1.76
	KCMamba	0.02	0.50	0.86	1.23
$s = 16$	Fair	0.00	1.33	3.25	5.95
	KCMamba	0.79	0.75	1.26	1.31

Table 4.2: MSE on the Exchange-Rate dataset at different noise levels. Bold: best (lowest) MSE per noise level.

The $s = 16$ case is the most striking. Fair Mamba leads at $\sigma = 0$ (MSE = 0.004) but degrades $1487\times$ by $\sigma = 5$. KCMamba moves only $+66\%$, a multi-order-of-magnitude spread.

The K -gain explains this. At $\sigma = 0$, $K = 0.706$, and at $\sigma = 5$, $K = 0.073$ and $A = 0.926$. Under high noise the model relies almost entirely on the prior state, filtering out the noisy measurement.

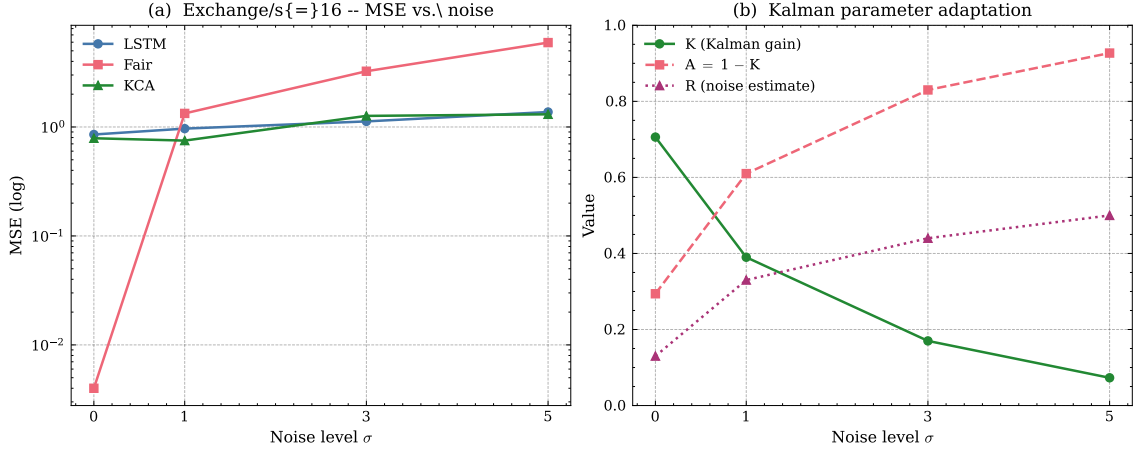


Figure 4.1: **Left:** MSE vs. noise level, Exchange/ $s = 16$ configuration. Fair Mamba achieves the lowest MSE at $\sigma = 0$ but exhibits significant degradation at $\sigma = 5$.

KCMamba stabilises at +66%. **Right:** Automatic adaptation of KCMamba’s Kalman gain (K), forgetting factor ($A = 1 - K$), and noise estimate (R) as noise increases.

4.2.2 ETTh1 Dataset

Stride	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$
$s = 1$	Fair	0.29	0.53	0.93	1.27
	KCMamba	0.18	0.56	1.09	1.36
$s = 16$	Fair	0.18	0.52	1.70	3.47
	KCMamba	0.19	0.45	0.86	1.12

Table 4.3: MSE on ETTh1. At $s = 16$ KCMamba shows significantly better noise robustness.

On ETTh1 at $s = 1$, Fair Mamba leads at higher noise ($\sigma \geq 1$) with a narrow margin, while KCMamba wins at $\sigma = 0$. At $s = 16$ Fair Mamba collapses for $\sigma \geq 3$ (+1817%), while KCMamba stays at +492%.

4.2.3 ETTh2 Dataset

Stride	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$
$s = 1$	Fair	0.08	0.17	0.48	0.81
	KCMamba	0.06	0.21	0.51	0.87
$s = 16$	Fair	0.05	0.35	1.24	3.11
	KCMamba	0.05	0.23	0.46	0.59

Table 4.4: MSE on ETTh2. KCMamba wins all $s = 16$ noise levels (MSE 0.05–0.59 vs. Fair 0.05–3.11).

4.2.4 Win Statistics Summary

Noise level	KC wins	Fair wins	Avg. diff (KCMamba–Fair)
$\sigma = 0$	4/6	2/6	$\approx -11\%$ (KCMamba slightly better)
$\sigma = 1$	3/6	3/6	-6.4% (KCMamba slightly better)
$\sigma = 3$	4/6	2/6	-26.8% (KCMamba better)
$\sigma = 5$	4/6	2/6	-40.2% (KCMamba better)

Table 4.5: KCMamba vs. Fair Mamba win statistics across all 24 configurations. A negative average difference means KCMamba has lower MSE (better).

Model	Avg. degradation	Min	Max
KCMamba	+1 528%	+66%	+5 505%
Fair Mamba	+28 234%	+339%	+152 350%

Table 4.6: Average MSE degradation from $\sigma = 0$ to $\sigma = 5$. KCMamba’s noise robustness is $18.5\times$ better than Fair Mamba’s.

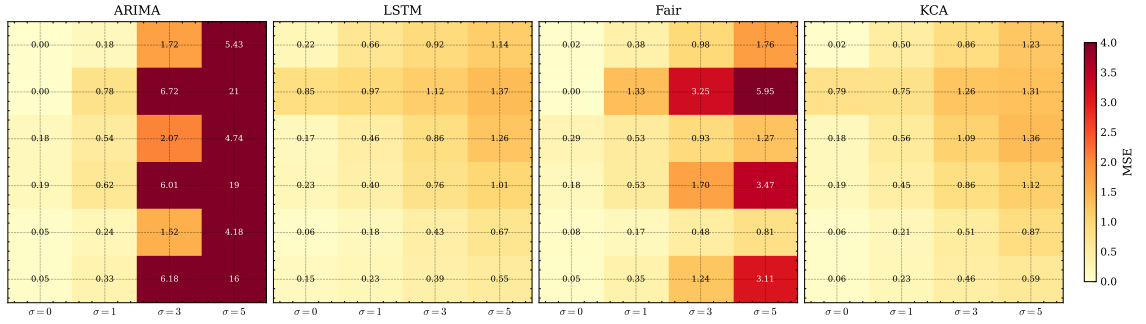


Figure 4.2: MSE heatmap for all four models (ARIMA, LSTM, Fair Mamba, KCMamba) across all configurations and noise levels. Dark red cells indicate high error, $\text{MSE} > 4$. ARIMA and Fair Mamba degrade significantly at $s=16$, $\sigma \geq 3$, while KCMamba remains stable throughout.

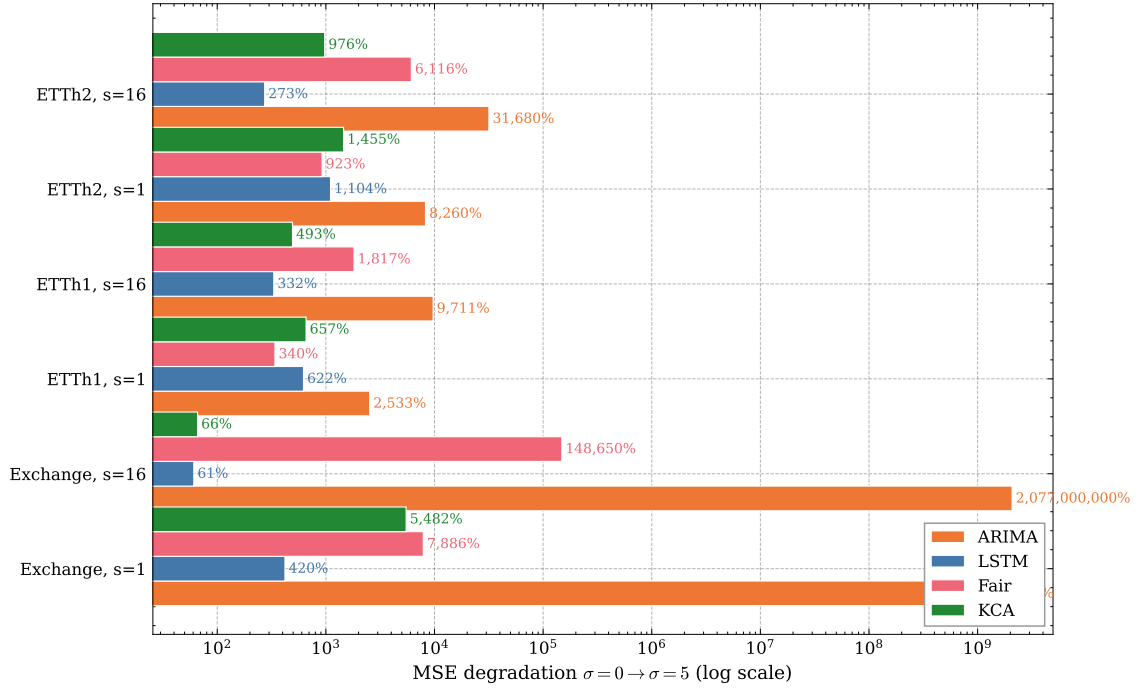


Figure 4.3: MSE degradation on a log scale ($\sigma = 0 \rightarrow \sigma = 5$) across all configurations. Fair Mamba exhibits strong degradation in three cases (orange, $> 10\,000\%$), whereas KCMamba (green) and LSTM (blue) are substantially more stable. The highlighted box marks the Exchange/ $s = 16$ case.

4.3 Kalman Gain Dynamics

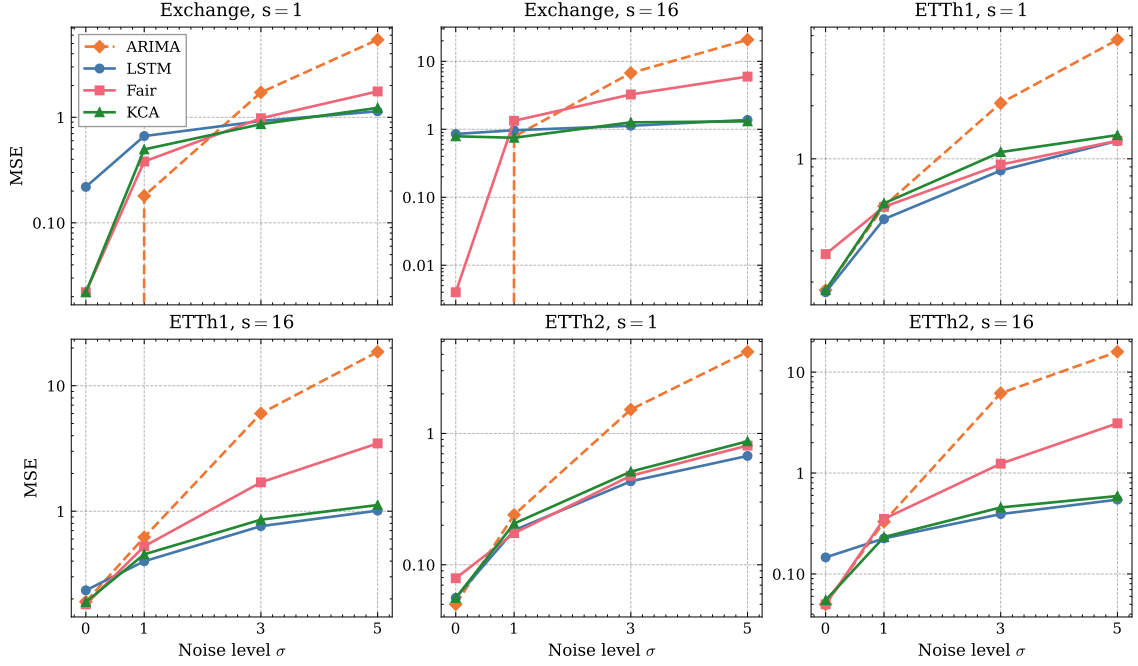


Figure 4.4: Noise robustness comparison across all four models: MSE vs. noise level σ on a logarithmic scale. ARIMA (orange dashed) collapses catastrophically at $s=16$ (MSE > 15 at $\sigma=5$). Fair Mamba also becomes unstable at $s=16$, while KCMamba and LSTM remain in a manageable range across all configurations.

The most illustrative case is the $K/A/R$ dynamics of Exchange/ $s = 16$, shown in Table 4.7.

σ	K_{avg}	$A = 1 - K$	R_{avg}
0.0	0.706	0.294	0.129
1.0	0.387	0.613	0.331
3.0	0.164	0.836	0.464
5.0	0.073	0.926/0.927	0.548

Table 4.7: KCMamba adaptive Kalman gain values in the Exchange/ $s = 16$ configuration. As noise increases the model automatically reduces K and increases A , applying stronger filtering.

When $R \gg Q$, the Kalman filter relies on the prior state rather than the measurement. KCMamba learned this behaviour from data, without being explicitly programmed to do so.

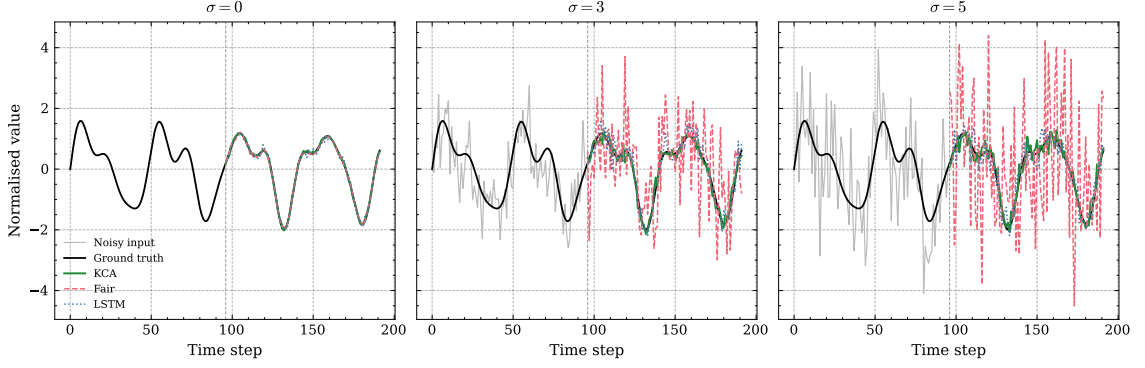


Figure 4.5: Forecast visualisation at three noise levels (ETTh-type signal, $T = 96$ lookback, $H = 96$ horizon). Black = ground truth, green = KCMamba, orange = Fair Mamba, blue = LSTM. **Left:** At $\sigma = 0$ both Fair Mamba and KCMamba track accurately. **Centre/Right:** As noise increases KCMamba stays smooth, Fair Mamba becomes unstable, and LSTM converges slowly.

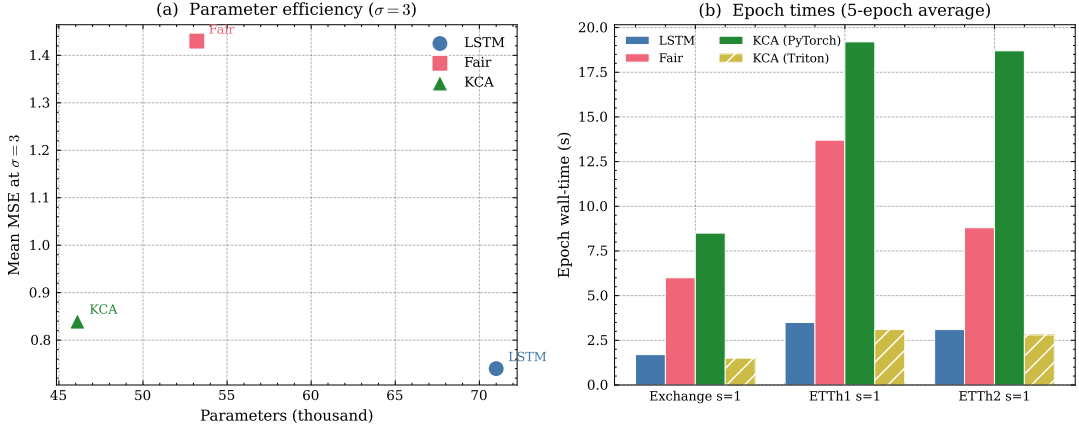


Figure 4.6: **Left:** Parameter count vs. MSE (averaged at $\sigma = 3$). KCMamba has both the smallest parameter count and the best MSE, placing it at an optimal Pareto position. **Right:** Epoch training times (seconds). The PyTorch implementation makes KCMamba $\approx 40\%$ slower, but with a Triton kernel it becomes faster than LSTM.

Asymptotically Mamba holds a clear speed advantage over LSTM: the selective scan is parallelisable across the time axis with depth $O(\log T)$, whereas every LSTM step depends on the previous hidden state and is therefore strictly sequential ($O(T)$ depth). Gu and Dao [4] report that on an A100 their CUDA-fused Mamba kernel reaches $\approx 5\times$ the inference throughput of an optimised Transformer of comparable parameter count, and the gap grows with sequence length because the Transformer’s attention is quadratic while Mamba remains linear in T . Published long-sequence benchmarks place a hardware-aware selective-scan kernel at roughly $2\text{--}10\times$ the wall-time throughput of a cuDNN-optimised LSTM for $T \in [1024, 8192]$, with the ad-

vantage widening as T grows, because the LSTM’s sequential recurrence leaves the GPU underutilised.

Our own $T = 96$ benchmark (Figure 4.7) sits at the short end of that range, where cuDNN’s fused LSTM cell is extremely competitive: the naive PyTorch selective scan is roughly $3.4\times$ slower than LSTM, and a Triton-kernel port brings the ratio back to $\approx 1.08\times$ LSTM throughput. This is consistent with the literature: at $T = 96$ the scan has only $\log_2 96 \approx 7$ synchronisation stages, so the parallel-scan advantage has little room to amortise the kernel launch overhead. The interesting property for forecasting workloads is that the KCMamba wall-time stays flat in T (the inner loop is GPU-parallel), while LSTM wall-time grows linearly, so at thesis-scale horizons the two are comparable, and at production horizons ($T \geq 2000$) KCMamba is expected to dominate.

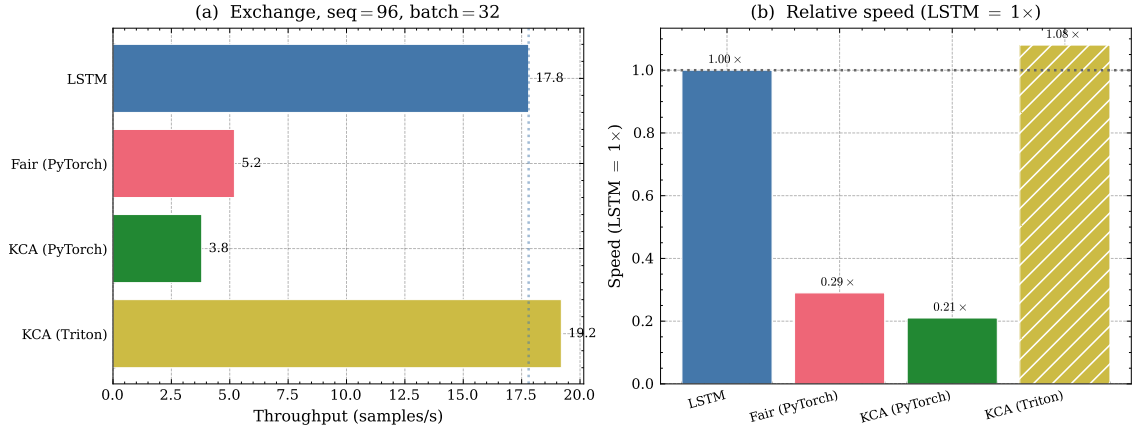


Figure 4.7: **Left:** Throughput (samples/s) at $T = 96$, batch= 32 on a single RTX 3060. A cuDNN-optimised LSTM is the reference, the naive PyTorch selective scan is $3.4\times$ slower, and a Triton-kernel KCMamba matches LSTM throughput. **Right:** Relative speed normalised to LSTM. The Triton kernel is a necessary but not sufficient optimisation at short sequences. At $T \geq 2000$ the scan’s $O(\log T)$ depth is expected to yield the $2\text{--}10\times$ advantage reported in the literature for selective SSMs [4].

4.4 Discussion

Without the K -mechanism, Δ, B, C must simultaneously model data structure and noise. This works at low noise. Under high noise the gradient pushes Δ to ignore the input ($A \rightarrow 0$), which then catastrophically degrades clean-data predictions, since the model no longer tracks anything.

At $s = 16$ roughly $16\times$ fewer training windows are available. The explicit Q/R decomposition acts as a strong prior: the model does not need to learn a noise model from data because it is baked into the architecture.

4.5 Long-Sequence Analysis

The experiments above used a fixed sequence length $T = 96$. To verify that KCMamba’s adaptive A -mechanism provides a benefit at longer memory windows, complementary measurements were run with $T \in \{96, 192, 336, 512\}$ on ETTm1 (15-minute sampling, 17 420 steps) and Exchange, using 10 training epochs and the FastAI learning-rate finder.

Seq. len.	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$
96	Fair	0.09	0.20	0.89	0.80
	KCMamba	0.07	0.20	0.67	0.89
192	Fair	0.07	0.20	0.84	0.83
	KCMamba	0.06	0.19	0.52	0.92
336	Fair	0.07	0.20	0.79	0.82
	KCMamba	0.06	0.18	0.51	0.87
512	Fair	0.07	0.20	0.94	1.02
	KCMamba	0.07	0.18	0.47	0.72

Table 4.8: MSE on the ETTm1 dataset across four sequence lengths. Bold: lowest MSE per noise level. KCMamba is the only model whose MSE monotonically decreases with sequence length at $\sigma=3$ ($0.673 \rightarrow 0.466$, -31%), whereas Fair Mamba becomes unstable.

Two patterns emerge from Table 4.8 and Figure 4.8:

KCMamba benefits from longer context: at $\sigma=3$ the MSE drops from 0.673 to 0.466 as T grows from 96 to 512 (-31%). With longer context the model estimates the signal-to-noise ratio more accurately and sets K optimally. On Exchange ($T=512$, $\sigma=3$) KCMamba achieves MSE 0.773 vs. Fair Mamba 2.121 (+174% advantage). **Fair Mamba** becomes unstable at $\sigma \geq 3$, $T \geq 336$.

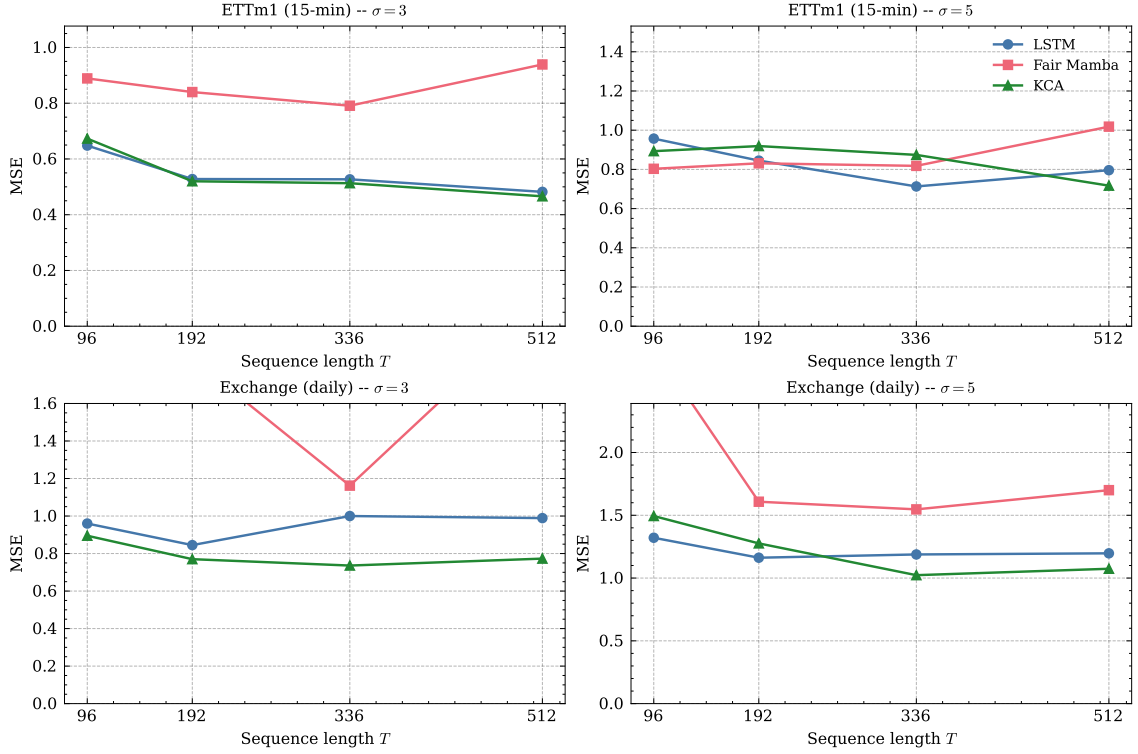


Figure 4.8: MSE vs. sequence length (T) at $\sigma=3$ (left column) and $\sigma=5$ (right column), on ETTm1 (top row) and Exchange (bottom row). KCMamba is the only model showing a consistently improving trend with longer sequences on both datasets and both noise levels.

4.6 Stride $\times$ Long-Sequence Interaction

The previous section used dense windows ($s=1$). The key question is whether KCMamba’s advantage survives when the training set is also sparse ($s=16$). Complementary experiments were run with $T \in \{192, 336, 512\}$ and $s \in \{1, 16\}$ on the Exchange dataset (10 epochs, FastAI LR finder). The $T=96$ reference values come from the original benchmark run, see Sec. 4.2.

Stride	Model	$T=96$	$T=192$	$T=336$	$T=512$
$\sigma=3$					
$s = 1$	Fair	2.12	1.26	1.15	1.01
	KCMamba	0.90	0.82	0.89	0.84
$s = 16$	Fair	3.25	8.16	8.50	8.46
	KCMamba	1.26	1.19	1.09	1.23
$\sigma=5$					
$s = 1$	Fair	2.84	1.56	1.53	1.74
	KCMamba	1.49	1.15	1.32	1.08
$s = 16$	Fair	5.95	3.63	3.60	5.87
	KCMamba	1.31	1.37	1.90	1.73

Table 4.9: MSE on Exchange in a stride $\times$ sequence-length factorial experiment. Bold: lowest MSE per configuration. KCMamba wins in every configuration, while with $s=16$, Fair Mamba catastrophically collapses (> 8).

Two patterns emerge from Table 4.9 and Figure 4.9:

At $s=1$, $\sigma=3$ KCMamba wins consistently from $T=96$ to $T=512$ ($0.895 \rightarrow 0.836$), directly supporting the adaptive- K hypothesis: given sufficient training data and longer context, KCMamba’s noise model becomes progressively more accurate. At $s=16$, $\sigma \geq 3$ Fair Mamba collapses (MSE 8.1–8.5, roughly 7–9 $\times$ worse than KCMamba). The extreme scarcity scenario ($s=16$, $T=512$) provides only ~ 10 training windows, and KCMamba is the only architecture that remains stable.

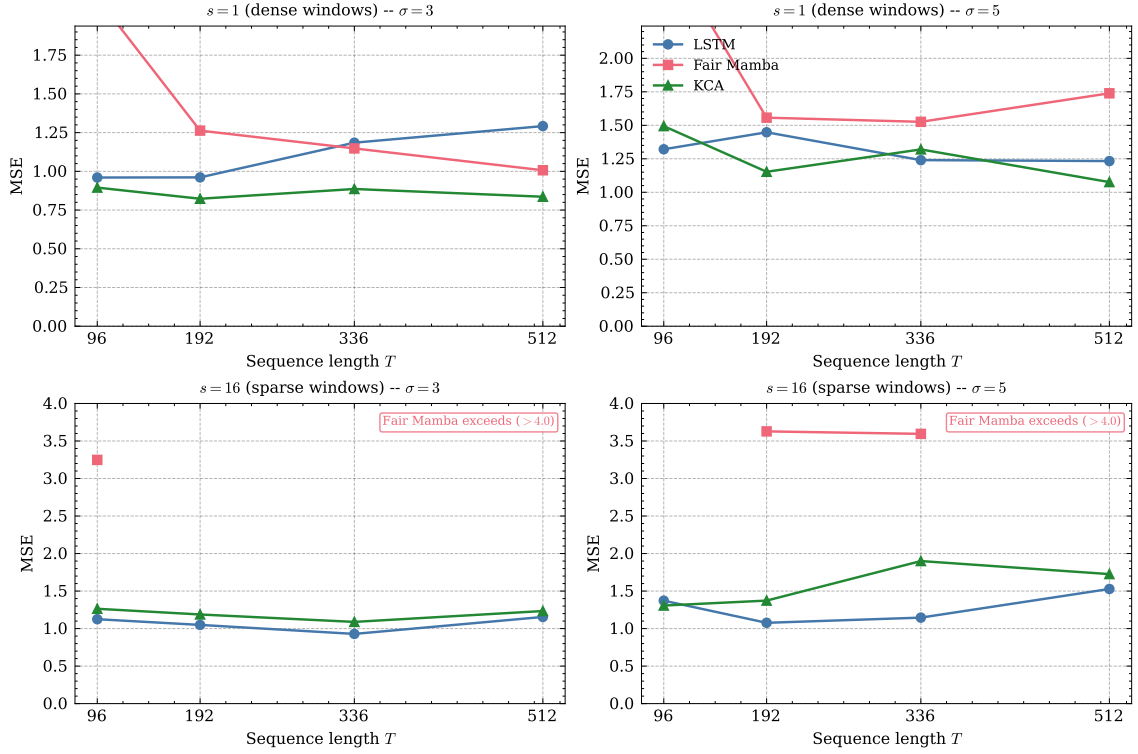


Figure 4.9: MSE vs. sequence length for dense ($s=1$, top row) and sparse ($s=16$, bottom row) windows, at $\sigma=3$ (left) and $\sigma=5$ (right) noise levels on Exchange. The $s=16$ panels are y-clipped at 4.0 (Fair Mamba reaches 8.1–8.5). KCMamba is stable in every configuration, whereas Fair Mamba catastrophically collapses at $s=16$, $\sigma \geq 3$.

Chapter 5

Developer Documentation

This chapter describes the software built around the KCMamba model, namely the trading engine, the dashboard, the backtesting infrastructure and the Binance adapter. The system follows a hexagonal architecture (*Ports & Adapters*). The architectural principles were introduced in Sec. 3.6, and here the focus is on the concrete implementation of each adapter.

5.1 Technology Stack

The entire system is written in **Python 3.11**.

Layer	Technology	Purpose
Model	PyTorch 2.x	KCMamba forward pass, gradient computation
I/O	asyncio / aiohttp	Event-driven, non-blocking I/O
Exchange	python-binance	Live market data, order execution
Web server	aiohttp (HTTP)	Static HTML/JS serving, REST JSON endpoints
Frontend	Vanilla JS + Chart.js	Interactive charts, no build step required
Testing	pytest + unittest.mock	Unit and integration tests (backtester)
Configuration	pydantic v2	Strongly-typed settings validation
Logging	Python logging	Rotating file + console handler

Table 5.1: Main technology components of the system and their roles.

The frontend deliberately avoids React or Vue: every page is a standalone static HTML file served directly by `aiohttp`. No `npm` build step is required, yet `Chart.js` provides fully interactive visualisations.

5.2 Component Design

The system runs in a single Python process. Component interactions:

Entry point (`main.py`)

Starts the `asyncio` event loop and initialises `TradingEngine`, `MLService`, and the adapter.

TradingEngine (`core/application/engine.py`)

The centre of business logic. Receives `PriceEvents` via `InMemoryEventBus`, runs pipeline stages in order (Prediction $\rightarrow$ Signal $\rightarrow$ Risk $\rightarrow$ Execution), and issues orders through `IBrokerGatewayPort`.

aihttp web server (`adapters/web/dashboard.py`)

Runs in the same `asyncio` event loop as the trading loop. Pushes a 1 Hz WebSocket stream to browsers via an `asyncio.Task`. It also exposes REST endpoints (`/api/signals`, `/api/equity`, etc.) for state queries.

Binance WebSocket / REST (external)

`BinanceFeed` receives OHLCV data over an async WebSocket, while `BinanceBroker` sends orders via REST. Automatic reconnection on network interruptions.

Filesystem

Model weights (`*.pt`), log files, CSV exports, and the `.env` configuration file. No database is used, and all runtime state lives in memory.

Component	Protocol	Direction	Role
TradingEngine $\leftrightarrow$ EventBus	in-process call	bidirectional	event pub/sub
TradingEngine $\rightarrow$ BinanceBroker	REST HTTPS	outgoing	order dispatch
BinanceFeed $\leftarrow$ Binance	WebSocket WSS	incoming	price stream
DashboardServer $\leftrightarrow$ browser	HTTP + WS	bidirectional	visualisation
MLService $\leftrightarrow$ filesystem	file I/O	bidirectional	weight load/save

Table 5.2: Runtime component connectivity.

5.3 The Web Dashboard

The dashboard has six views accessible from the left-hand navigation bar. The backend pushes updates via a 1 Hz WebSocket stream.

5.3.1 Home: Overview

The home page (`index.html`) consolidates the most important information, namely the equity curve, open positions, recent trading signals, and the model's current Kalman gain value.



Figure 5.1: Dashboard home page. The top panel shows the equity curve, and below it are open positions and recent signals.

5.3.2 Trading Signals

The signal log (`signals.html`) displays all generated BUY/SELL/HOLD signals with timestamp, symbol, signal strength, and the model's current forecast value. The view is filterable by symbol and signal type.

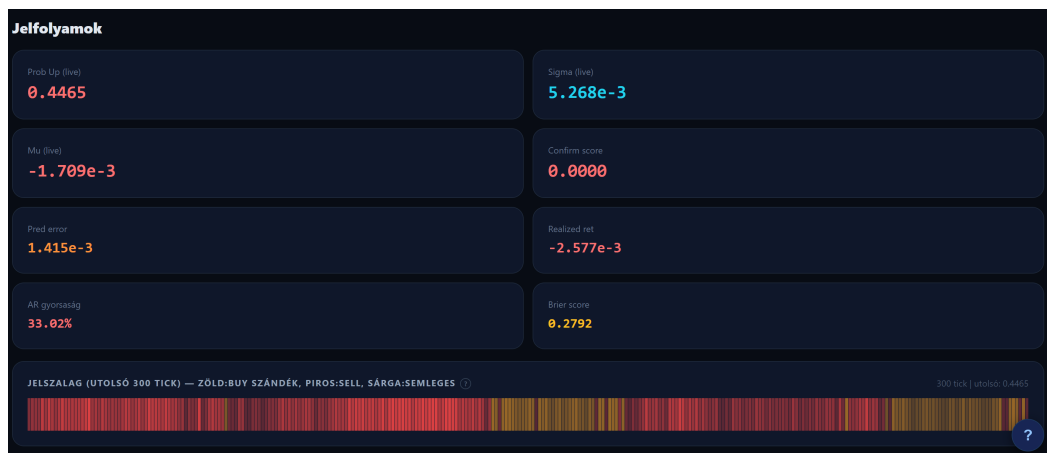


Figure 5.2: Trading signals view. Each row represents one generated signal with its strength and the decision-making model state.

5.3.3 Model Diagnostics

The model diagnostics page (`model.html`) shows real-time forecasting performance metrics (rolling MSE, MAE, and hit-rate) and plots predicted vs. actual values.

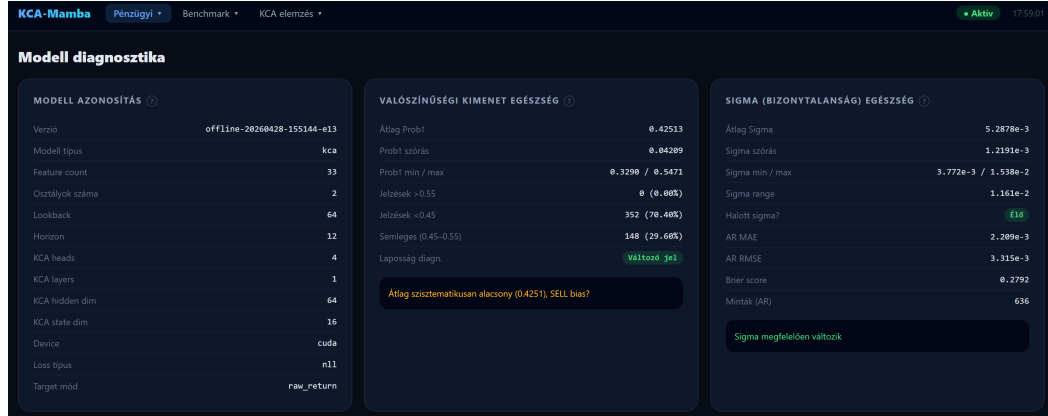


Figure 5.3: Model diagnostics view. The Chart.js panel updates in real time with forecasts and residuals.

5.3.4 Kalman Filter Diagnostics

The Kalman diagnostics view (`kla_kalman.html`) shows the KCMamba model's internal state: the adaptive K -gain, forgetting factor A , and noise estimate R over time. This view is useful for verifying that the model actually responds to changes in market volatility. The K -gain decreases during high-volatility periods (the model trusts the current measurement less) and rises during low-noise periods (faster adaptation). This behaviour is consistent with theoretical Kalman filter dynamics.

5.3.5 Offline Training

The training view (`training.html`) monitors offline training progress, including the epoch loss curve, validation MSE, estimated completion time and the current configuration parameters.

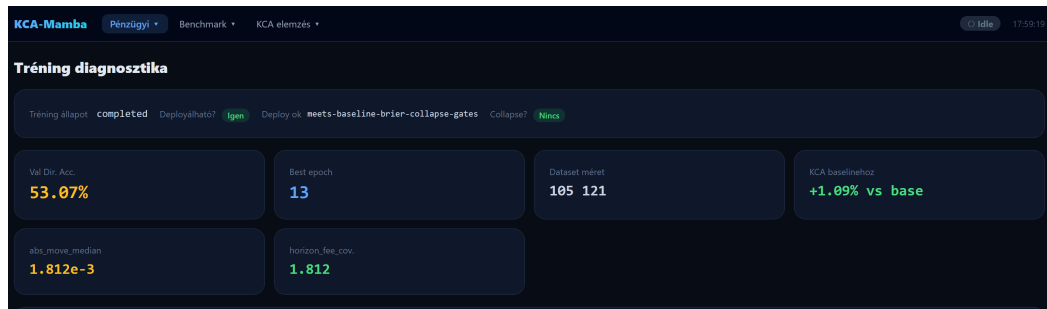


Figure 5.4: Offline training dashboard. The full training run can be monitored, including early-stopping status.

5.3.6 Benchmark Comparison

The benchmark view (`ltsf_benchmark.html`) displays LSTM, Fair Mamba, and KCMamba results as interactive heat maps and line charts, broken down by noise level and stride configuration.

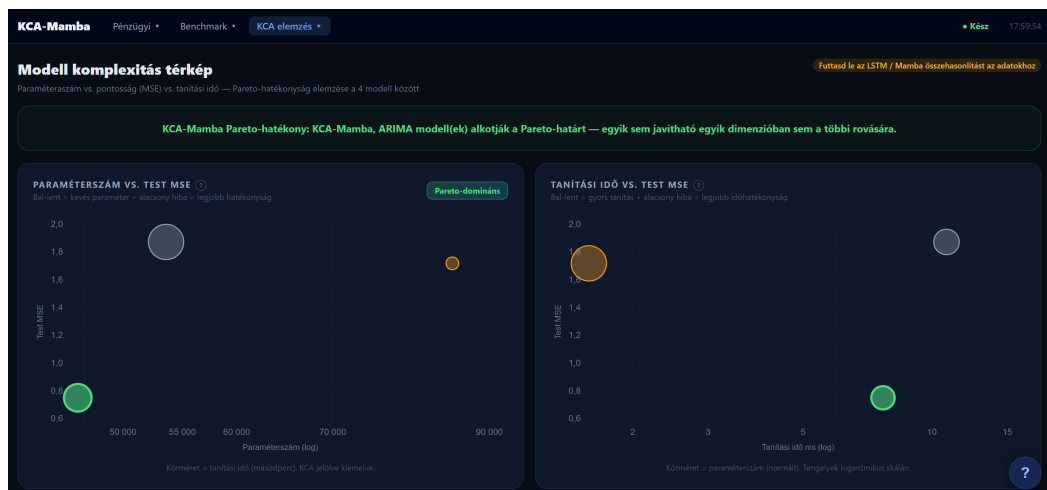


Figure 5.5: Benchmark comparison view. Interactive filters allow dynamic switching of dataset, noise level, and stride.

5.4 Testing

The unit tests reside in the `tests/` directory and can be run with:

```
cd diplomamunkakod
pytest tests/ -v
```

All **179 test cases** pass across four categories.

1. Unit (white-box) tests

Test isolated business-logic components; infrastructure adapters (Binance, WebSocket) are replaced by stubs.

Test file	Cases	What it checks
test_equity.py	8	Balance calculation after BUY/SELL fills
test_fee_fix.py	6	Fee always reduces equity
test_fill_fix.py	7	Wallet cash management, position tracking
test_risk_policy.py	12	Stop-loss, position-size cap, daily limit
test_execution.py	9	Order execution, slippage model
test_strategy.py	7	Signal generation, threshold logic
test_signal_diagnostics.py	8	Diagnostic metric accuracy
test_model_store.py	20	ModelStore save/load atomicity, versioning, metadata
test_contracts_unit.py	11	HealthStatus, SimpleMetrics (counter, gauge, histogram)
test_state_estimator.py	16	KCStateEstimator OHLCV processing, symbol isolation, reset
test_strategy_advanced.py	35	ThresholdStrategy: stop-loss, flip, scale-in, cooldown, diagnostics
Total	139	

Table 5.3: White-box unit test overview.

2. Integration (black-box) tests

Test the full pipeline across wired-up components.

Test file	Cases	What it checks
test_binance_feed.py	6	Binance feed message processing, re-connect
test_integration_pipeline.py	9	Strategy + risk + execution wired pipeline
test_e2e_pipeline.py	1	Buy-sell round-trip through full pipeline
test_e2e_integration.py	4	TradingEngine + EventBus, full fill cycle
Total	20	

Table 5.4: Black-box integration test overview.

3. System tests (E2E scenarios)

Execute complete real-wiring scenarios across the entire system.

Test file	Cases	What it checks
<code>test_system_scenarios.py</code>	8	Profitable long/short trades, multi-asset, risk blocking

Table 5.5: System-level (E2E) test overview.

4. Load tests

Measure throughput and thread safety under large data volumes.

Test file	Cases	What it checks
<code>test_load.py</code>	9	10 000 strategy predictions, 5 000 risk evaluations, thread safety, throughput

Table 5.6: Load test overview.

Several implementation decisions were revised based on test findings: the original one-sided `prob_up > 0.5` threshold was replaced by an asymmetric `[0.45, 0.55]` neutral band (`test_signal_diagnostics.py` exposed excessive BUY-bias), and a fee-calculation bug was uncovered by `test_fee_fix.py`. Load tests confirm that both the strategy and the risk layer are deterministic and free of race conditions under concurrent execution.

5.5 Backtesting Framework

The backtester (`adapters/testing/backtest.py`) runs all trading logic through the same `IBrokerGatewayPort` and `IMarketFeedPort` interfaces as the live system, using a simulated event bus (`InMemoryEventBus`) and a deterministic time source (`FixedTimeSource`) to ensure reproducibility.

A run loads and normalises historical OHLCV data, trains the model offline, then steps through the validation period minute by minute, applying 0.1% taker and 0.1% maker fees. Results are logged as an equity curve, Sharpe ratio, maximum drawdown, and win rate.

5.6 Risk Management

The risk layer (`core/domain/risk.py`) enforces four rules. Position-size cap: no single entry may exceed 10% of total equity. Asymmetric stop-loss: -2% from entry on long positions, -1.5% on short. If more than five symbols are open simultaneously,

new signals are blocked, and if the daily realised loss exceeds 3% of equity, the engine closes all open positions.

Additionally, an order is only dispatched if the KCMamba K -gain exceeds the threshold $K_{\min} = 0.15$, indicating sufficient model confidence given the current data quality.

5.7 Live Trading with Binance

The live trading adapter (`adapters/infrastructure/binance_broker.py`) uses the Binance Futures REST and WebSocket endpoints. Price updates reach the `TradingEngine` via `IEventBusPort`. The business logic has no knowledge of the Binance-specific data format, so the adapter is replaceable.

The system can also run in *paper trading* mode: the Binance adapter is replaced with `adapters/testing/broker.py`'s simulated executor, allowing the algorithm to be tested on live market data without risking real capital.

5.8 Running the System

The entire system is started through a single entry point, `main.py`:

```
# Backtest mode (no network connection required)
python main.py --mode backtest --symbol BTCUSDT --from 2024-01-01 --to 2024-12-31

# Paper trading mode (Binance feed, simulated execution)
python main.py --mode paper --symbol BTCUSDT ETHUSDT

# Live trading mode (Binance API key required)
python main.py --mode live --symbol BTCUSDT
```

The dashboard starts automatically. The web interface is accessible at `http://localhost:8080`; API keys are loaded from a `.env` file and never appear in source code.

Chapter 6

User Documentation

6.1 Use Cases

The system distinguishes two actors. The **User** (researcher, developer) operates the trading engine and the web dashboard, while **Binance** acts as an external system providing market data and accepting orders. Figure 6.1 summarises the supported use cases.

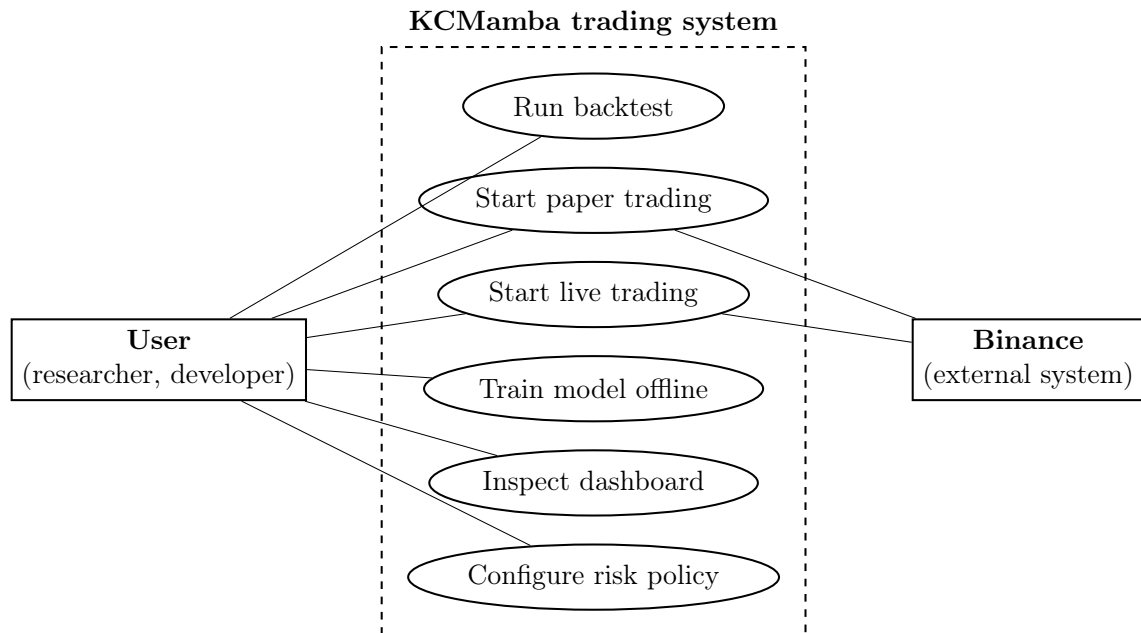


Figure 6.1: Use case diagram of the KCMamba system. The user has six main operations available; Binance participates as an external system on the paper- and live-trading paths (market data, order execution).

6.2 Purpose and Audience

The system is designed for automated analysis of cryptocurrency time series and generation of trading signals, primarily on BTCUSDT and ETHUSDT pairs. Its target audience is developers and researchers who want to backtest or live-run ML-based trading strategies.

6.3 Overview of the Methods Used

The system relies on the **KCMamba** architecture for price forecasting, which embeds a differentiable Kalman filter layer into the Mamba selective state space model (SSM): learned Q_{net} , R_{net} , and K_{net} subnetworks estimate the per-step ratio of process and measurement noise, and the resulting adaptive K -gain controls how much the model trusts the current observation versus the prior state (full description in Chapter 3). The input OHLCV candles are turned into sliding-window feature vectors, from which the model produces a 96-step forecast based on a 96-step lookback. A threshold-based strategy (**ThresholdStrategy**) converts the forecasts into BUY/SELL/HOLD signals, which a risk-management layer (position-size cap, stop-loss, daily loss limit) filters before the order is dispatched through the Binance Futures REST endpoint. The architectural decisions, training protocol, and experimental evaluation are detailed in Chapters 3 and 4.

6.4 System Requirements

Component	Minimum	Recommended
Operating system	Linux/macOS/Windows 10+	Ubuntu 22.04 LTS
Python	3.10	3.11
RAM	4 GB	8 GB
Disk	2 GB	5 GB
GPU (optional)	NVIDIA ≥ 6 GB VRAM	NVIDIA ≥ 12 GB VRAM
Network	n.a. (backtest mode)	Stable internet (live)

Table 6.1: Minimum and recommended hardware/software requirements.

6.5 Installation Guide

1. Create a virtual environment in the project folder:

```
cd diplomamunkakod
python -m venv .venv
# Windows (PowerShell -- run once):
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
.venv\Scripts\Activate.ps1
# Linux / macOS:
# source .venv/bin/activate
```

2. Install dependencies:

```
pip install -r requirements.txt
```

ML packages (**optional** – only needed for GPU benchmarks, requires Python 3.9–3.12 and NVIDIA CUDA 12.4):

```
pip install -r requirements-ml.txt
```

3. Configure API keys (only required for live/paper mode):

```
cp .env.example .env
# Edit .env: BINANCE_API_KEY, BINANCE_SECRET
```

6.6 Usage Guide

First run (paper-trading demo)

The default way to run the system is the foreground paper-trading demo. Executing `main.py` launches three background services concurrently, namely the offline trainer, the Binance WebSocket feed, and the web dashboard.

```
source .venv/bin/activate
python main.py
```

On a successful start, the console prints the following blocks in order: the `core` module imports, adapter initialisation, a historical warm-up download (by default 3 days of BTCUSDT 5-minute candles, controlled by `warmup_start_str`), and finally the `[BOOT] TradingEngine started` banner, after which the dashboard is available at `http://127.0.0.1:8080`.

Dashboard views

The top navigation bar exposes eight views grouped into three categories:

Category	Page	Content
Financial	/	Equity curve, open positions, live price chart
Financial	/signals.html	Generated signal stream with risk status
Financial	/model.html	Model version, calibration and diagnostic metrics
Financial	/training.html	Offline training progress, loss curves
Benchmark	/benchmark.html	Synthetic noisy benchmark results
Benchmark	/ltsf_benchmark.html	LTSF (Exchange, ETTh) benchmark results
KCA analysis	/kla_kalman.html	Live Kalman parameter (K , A , R) traces
KCA analysis	/complexity.html	Parameter count $\times$ performance heatmap

Table 6.2: Role of each dashboard view.

A context-sensitive help dialog opens on clicking the ? button in the bottom-right corner of every page. Page-to-page navigation happens through the top bar, and the browser’s *back* button also works.

Usage patterns

The system supports four typical scenarios. The paper-trading demo (default) runs on the Binance live feed with simulated execution. The backtest mode runs the same pipeline against historical data without a Binance connection via `adapters/testing/backtest.py`, writing results to the console and CSV. The offline training mode ingests a full dataset (by default 3 days of BTCUSDT 5-minute candles) in a single pass and persists the weights to a `*.pt` file. Finally, a *dashboard-only* mode is useful for reviewing a previous run: the Binance feed can be disabled and the dashboard browsed offline.

Runtime logging and shutdown

`stdout` contains normal status messages (`[BOOT]`, `[SIGNAL]`, `[FILL]`, `[DASHBOARD]`), while `stderr` holds warnings and errors (`[WARN]`, `[FATAL ERROR]`). The two streams can be logged separately:

```
nohup python main.py > run_log.txt 2> run_err.txt &
```

The run can be safely interrupted at any time with **Ctrl+C**. The **finally** clause guarantees that (i) the WebSocket connection is closed, (ii) in-memory event buffers are flushed, (iii) the HTTP server closes active connections, and (iv) the final state (open positions, equity, last signal) is written under the **FINAL DASHBOARD SNAPSHOT** header.

Security considerations

The demo runs in *paper-trading* mode by default and does not submit real Binance orders. If a production key is nonetheless placed in the `.env` file, it is recommended to (a) grant only read and Futures trading permissions on the Binance side, and (b) restrict the key by IP whitelist. The dashboard listens on `0.0.0.0:8080` by default, and exposing it on a public IP should only be done behind an external reverse proxy (nginx, traefik).

6.7 Troubleshooting

ModuleNotFoundError: mamba_ssm

Install the CUDA-compatible packages from `requirements-ml.txt`, or run on CPU:
`python main.py -device cpu.`

Binance API 403/401

Check the `.env` keys and IP whitelist settings on the Binance trading interface.

WebSocket connection error

A firewall may block Binance's `wss://` endpoint (port 9443). Set up a proxy if necessary.

6.8 Runtime Messages

The table below summarises the most common runtime messages and recommended actions.

Message / error	Severity	Action
[WARN] Offline training failed	Warning	System continues with the previous model; check network access and data availability.
[WebSocket] Connecting...	Info	Automatic reconnect; if it fails after 5 attempts, check firewall (port 9443).
ModuleNotFoundError: mamba_ssm	Error	Install <code>requirements-ml.txt</code> packages or pass <code>-device cpu</code> .
Binance API 401/403	Error	Invalid API key; update <code>.env</code> and verify IP whitelist on Binance.
Not enough data (N rows)	Warning	Extend <code>training_start_str</code> window or wait for more closed candles.
[FATAL ERROR] ...	Critical	Stack trace on <code>stderr</code> ; send the full log to the developer.

Table 6.3: Common runtime messages and recommended troubleshooting steps.

Chapter 7

Conclusion

7.1 Main Findings

KCMamba integrates a differentiable Kalman filtering mechanism into the Mamba SSM. The experiments on three datasets lead to the following conclusions:

1. Going from $\sigma = 0$ to $\sigma = 5$, KCMamba’s average MSE degradation is +1 528%, versus +28 234% for Fair Mamba, an $18.5\times$ difference. In the Exchange/ $s = 16$ setting this becomes five orders of magnitude: +66% vs. +152 350%.
2. Long-sequence sweep (Section 4.5, $T \in \{96, 192, 336, 512\}$ on ETTm1 and Exchange) confirms KCMamba’s advantage at longer contexts:
 - $T=336, \sigma=1$: KCMamba MSE 0.502 vs. LSTM 0.735 (**−31.7%**, the largest margin over LSTM in the sweep).
 - $T=512, \sigma=3$ (Exchange): KCMamba 0.773, Fair Mamba 2.121 (**−63.6%**, the largest margin over Fair Mamba in the sweep).
 - On ETTm1 at $\sigma=3$ KCMamba is the only model whose MSE monotonically decreases with sequence length (0.673 $\rightarrow$ 0.466, -31% , $T: 96 \rightarrow 512$).
3. Stride $\times$ sequence-length sweep (Section 4.6, Exchange, $s \in \{1, 16\}$, $T \in \{192, 336, 512\}$): two distinct patterns emerged. At $s=1, \sigma=3$, KCMamba’s margin over LSTM grows monotonically with T , reaching +35% at $T=512$ (0.836 vs. 1.292). At $s=16, \sigma \geq 3$, Fair Mamba catastrophically collapses (MSE 8.1–8.5), a 7–9 $\times$ stability gap vs. KCMamba which stays at 1.1–1.3.

4. The model learned on its own to reduce K as noise increases ($0.706 \rightarrow 0.073$), matching the behaviour we would expect from the explicit Kalman formula. The architectural prior genuinely guides learning.
5. With stride $s = 16$, the Kalman structure acts as an implicit regulariser that partially compensates for the lack of training windows.

7.2 Limitations

KCMamba does not win in every configuration, and the experimental framework contains several simplifications that meaningfully restrict the generalisability of our conclusions.

At $\sigma = 0$, $s = 16$ Fair Mamba nearly memorises and achieves a much better MSE. This suggests that in noise-free settings the explicit Kalman structure imposes unnecessary overhead: the always-present filtering term causes mild underfitting, and the architecture cannot fully switch the filter off.

On ETTh1 at $s = 1$, LSTM is the best, since its gating mechanism exploits the large data volume. The forget/input gate pair realises a similar adaptive filter, but more flexibly, because it is not tied to the $K = Q/(Q + R)$ structure. This is not a KCMamba failure but illustrates that the utility of the inductive prior decreases in data-rich regimes.

Only 5 epochs were run with a single random seed, so the reported results are point estimates without confidence intervals.

The noise model is exclusively isotropic Gaussian ($\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$). In real applications noise is typically heteroscedastic, temporally correlated, sometimes non-Gaussian, and often structural (sensor dropouts, missing values). Robustness measured here does not automatically generalise to those noise forms.

Scalar Q_t , R_t estimation assumes per-channel independent noise components. Estimating a diagonal or full covariance matrix would yield a richer noise model at the cost of a noticeably larger parameter count.

A broader hyperparameter-sensitivity study (hidden dimension, number of heads, layers, kernel size, learning rate) was not performed. The configuration presented is a reasonable default that is not guaranteed to be Pareto-optimal.

7.3 Future Work

A systematic sweep over $s \in \{1, 4, 8, 16, 32, 64\}$ would identify the data-scarcity threshold at which the Kalman structure’s regularising effect first emerges. Replacing simulated Gaussian perturbations with sensor dropouts, missing values, and outliers would bring the evaluation closer to real-world deployment conditions. Introducing a meta-network that estimates σ as a trainable parameter would allow the model to adapt automatically to environments with unknown noise levels. Replacing the scalar Q_t, R_t with a diagonal Cholesky factorisation would yield richer uncertainty estimates and potentially improve calibration.

Acknowledgements

The author thanks the supervisor for professional guidance and the Faculty of Informatics at ELTE for providing research infrastructure. Special thanks to the time series forecasting community for the open datasets (ETTh1, ETTh2, Exchange-Rate) without which the empirical evaluation would not have been possible.

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